

```
import { rcp } from '@kit.RemoteCommunicationKit';

const DEFAULT_RESUME_CWD: string = '~';
const OPENCODE_RESUMES_PATH: string = '/api/opencode/resumes';
const FILES_PATH: string = '/api/files';

export interface LoginRequest {
  baseUrl: string;
  registrationCode: string;
  deviceName: string;
  deviceId: string;
  publicKey: string;
}

export interface RegisterRequest {
  baseUrl: string;
  deviceId: string;
  publicKey: string;
  registrationCode: string;
  deviceName: string;
}

export interface RegisterResponse {
  activationToken: string;
  challenge: string;
  registrationCode?: string;
}

export interface ActivateRequest {
  baseUrl: string;
  deviceId: string;
  activationToken: string;
  signedChallenge: string;
}

export interface ChallengeRequest {
  baseUrl: string;
  deviceId: string;
}

export interface ChallengeResponse {
  nonce: string;
  expiresAt: string;
}

export interface VerifyRequest {
  baseUrl: string;
  deviceId: string;
  nonce: string;
  signature: string;
}
```

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}

export interface RefreshRequest {
  baseUrl: string;
  refreshToken: string;
}

export interface RefreshResponse {
  accessToken: string;
}

export interface LogoutRequest {
  baseUrl: string;
  refreshToken: string;
}

export interface LoginResponse {
  accessToken: string;
  refreshToken: string;
  sessionName: string;
  expiresIn: number;
  registrationCode?: string;
}

export interface ListResumesRequest {
  baseUrl: string;
  accessToken: string;
}

export interface StartResumeRequest {
  baseUrl: string;
  accessToken: string;
  cwd?: string;
}

export interface ResumeThreadRequest {
  baseUrl: string;
  accessToken: string;
  threadId: string;
}

export interface SendResumeMessageRequest {
  baseUrl: string;
  accessToken: string;
  threadId: string;
  message: string;
}

export interface ResumeMutationRequest {
  baseUrl: string;
}
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```
    accessToken: string;
    threadId: string;
}

export interface RenameResumeRequest extends ResumeMutationRequest {
    name: string;
}

export interface ListFilesRequest {
    baseUrl: string;
    accessToken: string;
}

export interface ServerFile {
    name: string;
    size: number;
    modifiedAt: string;
}

export interface DownloadFileRequest {
    baseUrl: string;
    accessToken: string;
    fileName: string;
}

export interface DownloadedFile {
    name: string;
    bytes: ArrayBuffer;
}

export interface ResumeItem {
    id: string;
    title: string;
    subtitle: string;
    status: string;
    updatedAt: string;
}

export interface ResumeSession {
    threadId: string;
    title: string;
    cwd: string;
    status: string;
    turns: ResumeTurn[];
}

export interface ResumeTurn {
    id: string;
    status: string;
    startedAt: string;
}
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    completedAt: string;
    items: ResumeContentItem[];
}

export interface ResumeContentItem {
    id: string;
    role: string;
    kind: string;
    text: string;
    status: string;
}

export interface GatewayClient {
    register(request: RegisterRequest): Promise<RegisterResponse>;
    activate(request: ActivateRequest): Promise<LoginResponse>;
    challenge(request: ChallengeRequest): Promise<ChallengeResponse>;
    verify(request: VerifyRequest): Promise<LoginResponse>;
    refresh(request: RefreshRequest): Promise<RefreshResponse>;
    logout(request: LogoutRequest): Promise<void>;
    checkHealth(baseUrl: string): Promise<boolean>;
    listResumes(request: ListResumesRequest): Promise<ResumeItem[]>;
    startResume(request: StartResumeRequest): Promise<ResumeSession>;
    resumeThread(request: ResumeThreadRequest): Promise<ResumeSession>;
    sendResumeMessage(request: SendResumeMessageRequest): Promise<ResumeSession>;
    renameResume(request: RenameResumeRequest): Promise<void>;
    deleteResume(request: ResumeMutationRequest): Promise<void>;
    listFiles(request: ListFilesRequest): Promise<ServerFile[]>;
    downloadFile(request: DownloadFileRequest): Promise<DownloadedFile>;
}

export interface RequestHeaders {
    contentType: string;
    authorization: string;
}

export interface RequestOptions {
    method: string;
    headers: RequestHeaders;
    body?: string;
}

export interface GatewayHttpResponse {
    statusCode: number;
    result: string;
}

export interface GatewayBinaryResponse {
    statusCode: number;
    body: ArrayBuffer;
}
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```
export interface GatewayTransport {
  request(url: string, options: RequestOptions): Promise<GatewayHttpResponse>;
  requestBinary(url: string, options: RequestOptions): Promise<GatewayBinaryResponse>;
}

export class GatewayHttpError extends Error {
  public statusCode: number;

  constructor(statusCode: number) {
    super(`Gateway request failed: ${statusCode}`);
    this.statusCode = statusCode;
  }
}

export class GatewayNetworkError extends Error {
  public detail: string;

  constructor(detail: string = '') {
    super('Gateway network request failed');
    this.detail = detail;
  }
}

export function isGatewayAuthenticationError(error: Object): boolean {
  if (!(error instanceof GatewayHttpError)) {
    return false;
  }

  return error.statusCode === 401 || error.statusCode === 403;
}

export function isGatewayNetworkError(error: Object): boolean {
  return error instanceof GatewayNetworkError;
}

interface LoginResponseBody {
  accessToken: string;
  refreshToken: string;
  sessionName: string;
  expiresIn: number;
  registrationCode?: string;
}

interface RegisterResponseBody {
  activationToken: string;
  challenge: string;
  registrationCode?: string;
}
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interface ChallengeResponseBody {
    nonce: string;
    expiresAt: string;
}

interface RefreshResponseBody {
    accessToken: string;
}

interface HealthResponseBody {
    status: string;
}

interface ListResumesResponseBody {
    items: ResumeItem[];
}

interface ListFilesResponseBody {
    items: ServerFile[];
}

interface ThreadStatus {
    type?: string;
}

interface ResumeThreadBodyThread {
    id?: string;
    name?: string;
    preview?: string;
    cwd?: string;
    status?: string | ThreadStatus;
    turns?: ResumeTurn[];
}

interface ResumeThreadResponseBody {
    threadId?: string;
    title?: string;
    cwd?: string;
    status?: string | ThreadStatus;
    turns?: ResumeTurn[];
    thread?: ResumeThreadBodyThread;
}

class SystemGatewayTransport implements GatewayTransport {
    private session: rcp.Session;
    private caContent: string;

    constructor(caContent: string = '') {
        this.caContent = caContent;
        const timeoutCfg: rcp.Timeout = {

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    connectMs: 30000,
    transferMs: 30000
  };
  if (caContent.length > 0) {
    console.info(`[SystemGatewayTransport] Creating session with CA cert, length=${caContent.length}`);
    const caCert: rcp.CertificateAuthority = { content: this.caContent };
    const sessionCfg: rcp.SessionConfiguration = {
      requestConfiguration: {
        transfer: {
          timeout: timeoutCfg
        },
        security: { remoteValidation: caCert }
      }
    };
    this.session = rcp.createSession(sessionCfg);
  } else {
    console.info('[SystemGatewayTransport] Creating session WITHOUT CA cert (default trust)');
    const sessionCfg: rcp.SessionConfiguration = {
      requestConfiguration: {
        transfer: {
          timeout: timeoutCfg
        }
      }
    };
    this.session = rcp.createSession(sessionCfg);
  }
}

async request(url: string, options: RequestOptions): Promise<GatewayHttpResponse> {
  const req = new rcp.Request(url, options.method);

  const headers: Record<string, string> = {};
  if (options.headers.contentType.length > 0) {
    headers['content-type'] = options.headers.contentType;
  }
  if (options.headers.authorization.length > 0) {
    headers['authorization'] = options.headers.authorization;
  }
  req.headers = headers;

  if (options.body !== undefined && options.body.length > 0) {
    req.content = options.body;
  }

  try {
    const response = await this.session.fetch(req);
    const result: string = JSON.stringify(response.toJSON());

    return {
      statusCode: response.statusCode,

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        result: result
    };
} catch (error) {
    const detail: string = typeof error === 'object' && error !== null ? JSON.stringify(error) : String(error);
    console.error(`[GatewayTransport] request failed: url=${url}, method=${options.method}, error=${detail}`);
    throw new GatewayNetworkError(detail);
}
}

async requestBinary(url: string, options: RequestOptions): Promise<GatewayBinaryResponse> {
    const req = new rcp.Request(url, options.method);

    const headers: Record<string, string> = {};
    if (options.headers.contentType.length > 0) {
        headers['content-type'] = options.headers.contentType;
    }
    if (options.headers.authorization.length > 0) {
        headers['authorization'] = options.headers.authorization;
    }
    req.headers = headers;

    try {
        const response = await this.session.fetch(req);
        return {
            statusCode: response.statusCode,
            body: response.body !== undefined ? response.body : new ArrayBuffer(0)
        };
    } catch (error) {
        const detail: string = typeof error === 'object' && error !== null ? JSON.stringify(error) : String(error);
        console.error(`[GatewayTransport] requestBinary failed: url=${url}, method=${options.method},
error=${detail}`);
        throw new GatewayNetworkError(detail);
    }
}

export class HttpGatewayClient implements GatewayClient {
    private transport: GatewayTransport;

    constructor(transport: GatewayTransport = new SystemGatewayTransport()) {
        this.transport = transport;
    }

    static createWithCert(certContent: string): HttpGatewayClient {
        return new HttpGatewayClient(new SystemGatewayTransport(certContent));
    }

    async register(request: RegisterRequest): Promise<RegisterResponse> {
        const response = await this.transport.request(joinUrl(request.baseUrl, '/api/auth/register'), {
            method: 'POST',
            headers: {

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        contentType: 'application/json',
        authorization: ''
    },
    body: JSON.stringify({
        deviceId: request.deviceId,
        publicKey: request.publicKey,
        registrationCode: request.registrationCode,
        deviceName: request.deviceName
    })
});
ensureSuccess(response);

const body = JSON.parse(response.result) as RegisterResponseBody;
return {
    activationToken: body.activationToken,
    challenge: body.challenge,
    registrationCode: body.registrationCode
};
}

async activate(request: ActivateRequest): Promise<LoginResponse> {
    const response = await this.transport.request(joinUrl(request.baseUrl, '/api/auth/activate'), {
        method: 'POST',
        headers: {
            contentType: 'application/json',
            authorization: ''
        },
        body: JSON.stringify({
            deviceId: request.deviceId,
            activationToken: request.activationToken,
            signedChallenge: request.signedChallenge
        })
    });
    ensureSuccess(response);

    const body = JSON.parse(response.result) as LoginResponseBody;
    return {
        accessToken: body.accessToken,
        refreshToken: body.refreshToken,
        sessionName: 'opencode-main',
        expiresIn: 900,
        registrationCode: body.registrationCode
    };
}

async challenge(request: ChallengeRequest): Promise<ChallengeResponse> {
    const response = await this.transport.request(joinUrl(request.baseUrl, '/api/auth/challenge'), {
        method: 'POST',
        headers: {
            contentType: 'application/json',

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        authorization: ''
    },
    body: JSON.stringify({
        deviceId: request.deviceId
    })
});
ensureSuccess(response);

const body = JSON.parse(response.result) as ChallengeResponseBody;
return {
    nonce: body.nonce,
    expiresAt: body.expiresAt
};
}

async verify(request: VerifyRequest): Promise<LoginResponse> {
    const response = await this.transport.request(joinUrl(request.baseUrl, '/api/auth/verify'), {
        method: 'POST',
        headers: {
            contentType: 'application/json',
            authorization: ''
        },
        body: JSON.stringify({
            deviceId: request.deviceId,
            nonce: request.nonce,
            signature: request.signature
        })
    });
    ensureSuccess(response);

    const body = JSON.parse(response.result) as LoginResponseBody;
    return {
        accessToken: body.accessToken,
        refreshToken: body.refreshToken,
        sessionName: 'opencode-main',
        expiresIn: 900
    };
}

async refresh(request: RefreshRequest): Promise<RefreshResponse> {
    const response = await this.transport.request(joinUrl(request.baseUrl, '/api/auth/refresh'), {
        method: 'POST',
        headers: {
            contentType: 'application/json',
            authorization: ''
        },
        body: JSON.stringify({
            refreshToken: request.refreshToken
        })
    });
};

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ensureSuccess(response);

const body = JSON.parse(response.result) as RefreshResponseBody;
return {
  accessToken: body.accessToken
};
}

async logout(request: LogoutRequest): Promise<void> {
  try {
    await this.transport.request(joinUrl(request.baseUrl, '/api/auth/logout'), {
      method: 'POST',
      headers: {
        contentType: 'application/json',
        authorization: ''
      },
      body: JSON.stringify({
        refreshToken: request.refreshToken
      })
    });
  } catch {
    // Best-effort
  }
}

async checkHealth(baseUrl: string): Promise<boolean> {
  const response = await this.transport.request(joinUrl(baseUrl, '/health'), {
    method: 'GET',
    headers: {
      contentType: '',
      authorization: ''
    }
  });
  if (response.statusCode !== 200) {
    return false;
  }
  try {
    const body = JSON.parse(response.result) as HealthResponseBody;
    return body.status === 'ok';
  } catch {
    return false;
  }
}

async listResumes(request: ListResumesRequest): Promise<ResumeItem[]> {
  const url = joinUrl(request.baseUrl, OPENCODE_RESUMES_PATH);
  console.info(`[listResumes] 请求 URL: ${url}`);
  const response = await this.transport.request(url, {
    method: 'GET',
    headers: {

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        contentType: '',
        authorization: `Bearer ${request.accessToken}`
    }
});
ensureSuccess(response);

const body = JSON.parse(response.result) as ListResumesResponseBody;
return body.items;
}

async startResume(request: StartResumeRequest): Promise<ResumeSession> {
    const response = await this.transport.request(joinUrl(request.baseUrl, OPENCODE_RESUMES_PATH), {
        method: 'POST',
        headers: {
            contentType: 'application/json',
            authorization: `Bearer ${request.accessToken}`
        },
        body: JSON.stringify({
            cwd: normalizeString(request.cwd)
        })
    });
    ensureSuccess(response);

    return toResumeSession(JSON.parse(response.result) as ResumeThreadResponseBody, '');
}

async resumeThread(request: ResumeThreadRequest): Promise<ResumeSession> {
    const response = await this.transport.request(
        joinUrl(request.baseUrl, `${OPENCODE_RESUMES_PATH}/${encodeURIComponent(request.threadId)}/resume`),
        {
            method: 'POST',
            headers: {
                contentType: 'application/json',
                authorization: `Bearer ${request.accessToken}`
            },
            body: JSON.stringify({
                threadId: request.threadId
            })
        }
    );
    ensureSuccess(response);

    return toResumeSession(JSON.parse(response.result) as ResumeThreadResponseBody, request.threadId);
}

async sendResumeMessage(request: SendResumeMessageRequest): Promise<ResumeSession> {
    const response = await this.transport.request(
        joinUrl(request.baseUrl, `${OPENCODE_RESUMES_PATH}/${encodeURIComponent(request.threadId)}/messages`),
        {
            method: 'POST',

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        headers: {
            contentType: 'application/json',
            authorization: `Bearer ${request.accessToken}`
        },
        body: JSON.stringify({
            message: request.message
        })
    }
);
ensureSuccess(response);

return toResumeSession(JSON.parse(response.result) as ResumeThreadResponseBody, request.threadId);
}

async renameResume(request: RenameResumeRequest): Promise<void> {
    const response = await this.transport.request(
        joinUrl(request.baseUrl, `${OPENCODE_RESUMES_PATH}/${encodeURIComponent(request.threadId)}/name`),
        {
            method: 'POST',
            headers: {
                contentType: 'application/json',
                authorization: `Bearer ${request.accessToken}`
            },
            body: JSON.stringify({
                name: request.name
            })
        }
    );
    ensureSuccess(response);
}

async deleteResume(request: ResumeMutationRequest): Promise<void> {
    const response = await this.transport.request(
        joinUrl(request.baseUrl, `${OPENCODE_RESUMES_PATH}/${encodeURIComponent(request.threadId)}`),
        {
            method: 'DELETE',
            headers: {
                contentType: '',
                authorization: `Bearer ${request.accessToken}`
            }
        }
    );
    ensureSuccess(response);
}

async listFiles(request: ListFilesRequest): Promise<ServerFile[]> {
    const url = joinUrl(request.baseUrl, FILES_PATH);
    console.info(`[listFiles] 请求 URL: ${url}`);
    const response = await this.transport.request(url, {
        method: 'GET',

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    headers: {
      contentType: '',
      authorization: `Bearer ${request.accessToken}`
    }
  });
  ensureSuccess(response);

  const body = JSON.parse(response.result) as ListFilesResponseBody;
  return body.items;
}

async downloadFile(request: DownloadFileRequest): Promise<DownloadedFile> {
  const url = joinUrl(request.baseUrl, `${FILES_PATH}/${encodeURIComponent(request.fileName)}/download`);
  console.info(`[downloadFile] 请求 URL: ${url}`);
  const response = await this.transport.requestBinary(url, {
    method: 'GET',
    headers: {
      contentType: '',
      authorization: `Bearer ${request.accessToken}`
    }
  });
  ensureSuccessCode(response.statusCode);

  return {
    name: request.fileName,
    bytes: response.body
  };
}

export class MockGatewayClient implements GatewayClient {
  async checkHealth(baseUrl: string): Promise<boolean> {
    return true;
  }

  async register(request: RegisterRequest): Promise<RegisterResponse> {
    return {
      activationToken: 'act-mock-001',
      challenge: 'mock-challenge'
    };
  }

  async activate(request: ActivateRequest): Promise<LoginResponse> {
    return {
      accessToken: 'router_access_mock',
      refreshToken: 'router_refresh_mock',
      sessionId: 'opencode-main',
      expiresIn: 900
    };
  }
}

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async challenge(request: ChallengeRequest): Promise<ChallengeResponse> {
  return {
    nonce: 'mock-nonce',
    expiresAt: new Date(Date.now() + 120000).toISOString()
  };
}

async verify(request: VerifyRequest): Promise<LoginResponse> {
  return {
    accessToken: 'router_access_mock',
    refreshToken: 'router_refresh_mock',
    sessionName: 'opencode-main',
    expiresIn: 900
  };
}

async refresh(request: RefreshRequest): Promise<RefreshResponse> {
  return {
    accessToken: 'router_access_mock_refreshed'
  };
}

async logout(request: LogoutRequest): Promise<void> {
}

async listResumes(request: ListResumesRequest): Promise<ResumeItem[]> {
  return [
    {
      id: 'thread-login-fix',
      title: 'Fix router login',
      subtitle: '/srv/projects/codex-router',
      status: 'completed',
      updatedAt: '2026-07-02T10:54:44.000Z'
    },
    {
      id: 'thread-backend',
      title: 'Bridge OpenCode app-server',
      subtitle: '/srv/projects/codex-router',
      status: 'running',
      updatedAt: '2026-07-02T11:20:00.000Z'
    }
  ];
}

async startResume(request: StartResumeRequest): Promise<ResumeSession> {
  return {
    threadId: 'thread-new',
    title: 'New OpenCode Resume',
    cwd: normalizeString(request.cwd) || DEFAULT_RESUME_CWD,
  };
}

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        status: 'running',
        turns: []
    };
}

async resumeThread(request: ResumeThreadRequest): Promise<ResumeSession> {
    return {
        threadId: request.threadId,
        title: request.threadId === 'thread-login-fix' ? 'Fix router login' : 'Bridge OpenCode app-server',
        cwd: '/srv/projects/codex-router',
        status: 'running',
        turns: [
            {
                id: 'turn-mock',
                status: 'completed',
                startedAt: '2026-07-02T10:54:44.000Z',
                completedAt: '2026-07-02T10:55:00.000Z',
                items: [
                    {
                        id: 'turn-mock:item-1',
                        role: 'user',
                        kind: 'message',
                        text: 'Open saved OpenCode session',
                        status: ''
                    },
                    {
                        id: 'turn-mock:item-2',
                        role: 'assistant',
                        kind: 'message',
                        text: 'Loaded session content from OpenCode.',
                        status: ''
                    }
                ]
            }
        ]
    };
}

async sendResumeMessage(request: SendResumeMessageRequest): Promise<ResumeSession> {
    const session = await this.resumeThread({
        baseUrl: request.baseUrl,
        accessToken: request.accessToken,
        threadId: request.threadId
    });
    return {
        threadId: session.threadId,
        title: session.title,
        cwd: session.cwd,
        status: 'running',
        turns: session.turns.concat([

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    {
      id: 'turn-mock-reply',
      status: 'completed',
      startedAt: '2026-07-02T11:00:00.000Z',
      completedAt: '2026-07-02T11:01:00.000Z',
      items: [
        {
          id: 'turn-mock-reply:item-1',
          role: 'user',
          kind: 'message',
          text: request.message,
          status: ''
        },
        {
          id: 'turn-mock-reply:item-2',
          role: 'assistant',
          kind: 'message',
          text: 'Mock reply from OpenCode.',
          status: ''
        }
      ]
    }
  ])
})
};
}

async renameResume(_request: RenameResumeRequest): Promise<void> {
}

async deleteResume(_request: ResumeMutationRequest): Promise<void> {
}

async listFiles(_request: ListFilesRequest): Promise<ServerFile[]> {
  return [
    {
      name: '红楼梦第一节课.pptx',
      size: 39610,
      modifiedAt: '2026-09-02T08:43:38.000Z'
    },
    {
      name: '调研报告.docx',
      size: 25410,
      modifiedAt: '2026-09-01T03:20:00.000Z'
    }
  ];
}

async downloadFile(_request: DownloadFileRequest): Promise<DownloadedFile> {
  return {
    name: _request.fileName,
  }
}

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        bytes: new ArrayBuffer(0)
    };
}
}

function toResumeSession(body: ResumeThreadResponseBody, fallbackThreadId: string): ResumeSession {
    const thread = body.thread;
    const status = body.status !== undefined ? body.status : thread?.status;
    const title = normalizeString(body.title) || normalizeString(thread?.name) || normalizeString(thread?.preview)
    ||
    fallbackThreadId;

    return {
        threadId: normalizeString(body.threadId) || normalizeString(thread?.id) || fallbackThreadId,
        title,
        cwd: normalizeString(body.cwd) || normalizeString(thread?.cwd),
        status: normalizeStatus(status),
        turns: normalizeTurns(body.turns !== undefined ? body.turns : thread?.turns)
    };
}

function normalizeTurns(value: ResumeTurn[] | undefined): ResumeTurn[] {
    if (!Array.isArray(value)) {
        return [];
    }

    return value.map((turn: ResumeTurn, turnIndex: number) => {
        const turnId = normalizeString(turn.id) || `turn-${turnIndex + 1}`;
        const normalized: ResumeTurn = {
            id: turnId,
            status: normalizeString(turn.status) || 'unknown',
            startedAt: normalizeString(turn.startedAt),
            completedAt: normalizeString(turn.completedAt),
            items: normalizeItems(turn.items, turnId)
        };
        return normalized;
    }).filter((turn: ResumeTurn) => turn.items.length > 0 || isActiveResumeTurnStatus(turn.status));
}

function normalizeItems(value: ResumeContentItem[], turnId: string): ResumeContentItem[] {
    if (!Array.isArray(value)) {
        return [];
    }

    return value.map((item: ResumeContentItem, itemIndex: number) => {
        const normalized: ResumeContentItem = {
            id: normalizeString(item.id) || `${turnId}:item-${itemIndex + 1}`,
            role: normalizeString(item.role) || 'system',
            kind: normalizeString(item.kind) || 'message',
            text: normalizeString(item.text),
            status: normalizeString(item.status)
        };
    });
}

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    };
    return normalized;
  }).filter((item: ResumeContentItem) => item.text.length > 0);
}

function normalizeString(value: string | undefined): string {
  if (typeof value !== 'string') {
    return '';
  }

  return value.trim();
}

function normalizeStatus(value: string | ThreadStatus | undefined): string {
  if (typeof value === 'string') {
    return value;
  }

  if (value && typeof value.type === 'string') {
    return value.type;
  }

  return 'unknown';
}

export function isActiveResumeTurnStatus(status: string): boolean {
  return status === 'inProgress' || status === 'running' || status === 'queued' || status === 'in_progress';
}

function joinUrl(baseUrl: string, path: string): string {
  return `${baseUrl.replace(/\/+$/, '')}${path}`;
}

function ensureSuccess(response: GatewayHttpResponse): void {
  if (response.statusCode < 200 || response.statusCode >= 300) {
    throw new GatewayHttpError(response.statusCode);
  }
}

function ensureSuccessCode(statusCode: number): void {
  if (statusCode < 200 || statusCode >= 300) {
    throw new GatewayHttpError(statusCode);
  }
}

// 固定服务器配置: App 只能连接该服务器, 地址不可修改、连接不可删除。
// 注意: 更改此处需重新构建并上架(证书固定 + 硬编码地址)。
export const DEFAULT_GATEWAY_SCHEME: string = 'https';
export const DEFAULT_GATEWAY_HOST: string = '8.153.174.88';
export const DEFAULT_GATEWAY_PORT: number = 8443;
export const DEFAULT_GATEWAY_DISPLAY: string = `${DEFAULT_GATEWAY_HOST}:${DEFAULT_GATEWAY_PORT}`;

```

```

export const DEFAULT_GATEWAY_NAME: string = 'AIRouter';
import { preferences } from '@kit.ArkData';
import { common } from '@kit.AbilityKit';

const DEVICE_ID_PREFS_NAME: string = 'ai_router_device';
const DEVICE_ID_PREFS_KEY: string = 'device_id';

let cachedDeviceId: string = '';

function generateUUID(): string {
  const chars = '0123456789abcdef';
  let uuid = '';
  for (let i = 0; i < 36; i++) {
    if (i === 8 || i === 13 || i === 18 || i === 23) {
      uuid += '-';
    } else {
      const randomIndex = Math.floor(Math.random() * 16);
      uuid += chars.charAt(randomIndex);
    }
  }
  return uuid;
}

export async function getDeviceId(context: common.Context): Promise<string> {
  if (cachedDeviceId.length > 0) {
    return cachedDeviceId;
  }

  try {
    const store = await preferences.getPreferences(context, DEVICE_ID_PREFS_NAME);
    const saved = await store.get(DEVICE_ID_PREFS_KEY, '');
    if (typeof saved === 'string' && saved.trim().length > 0) {
      cachedDeviceId = saved.trim();
      return cachedDeviceId;
    }
  } catch (error) {}

  const newId = generateUUID();
  cachedDeviceId = newId;

  try {
    const store = await preferences.getPreferences(context, DEVICE_ID_PREFS_NAME);
    await store.put(DEVICE_ID_PREFS_KEY, newId);
    await store.flush();
  } catch (error) {}

  return newId;
}

```

```

import { huks } from '@kit.UniversalKeystoreKit';
import { BusinessError } from '@kit.BasicServicesKit';

const KEY_ALIAS: string = 'ai_router_device_auth';

function stringToUint8Array(str: string): Uint8Array {
  const result: number[] = [];
  for (let i = 0; i < str.length; i++) {
    result.push(str.charCodeAt(i) & 0xff);
  }
  return new Uint8Array(result);
}

function uint8ArrayToBase64(data: Uint8Array): string {
  const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/';
  let result = '';
  for (let i = 0; i < data.length; i += 3) {
    const b0 = data[i];
    const b1 = i + 1 < data.length ? data[i + 1] : 0;
    const b2 = i + 2 < data.length ? data[i + 2] : 0;
    result += chars.charAt(b0 >> 2);
    result += chars.charAt(((b0 & 3) << 4) | (b1 >> 4));
    result += i + 1 < data.length ? chars.charAt(((b1 & 15) << 2) | (b2 >> 6)) : '=';
    result += i + 2 < data.length ? chars.charAt(b2 & 63) : '=';
  }
  return result;
}

function wrapDerInPem(derData: Uint8Array): string {
  const b64 = uint8ArrayToBase64(derData);
  const lines: string[] = [];
  for (let i = 0; i < b64.length; i += 64) {
    lines.push(b64.substring(i, i + 64));
  }
  return `-----BEGIN PUBLIC KEY-----\n${lines.join('\n')}\n-----END PUBLIC KEY-----`;
}

function buildEd25519SpkiDer(rawPublicKey: Uint8Array): Uint8Array {
  // Ed25519 OID: 1.3.101.112 = 0x2b 0x65 0x70
  // SPKI structure:
  // SEQUENCE {
  //   SEQUENCE { OID 1.3.101.112 }
  //   BIT STRING { 0x00 + rawPublicKey }
  // }

  let keyBytes = rawPublicKey;
  if (keyBytes.length === 33 && keyBytes[0] === 0x00) {
    keyBytes = keyBytes.slice(1);
  }
}

```

```

const algorithmId = new Uint8Array([0x30, 0x05, 0x06, 0x03, 0x2b, 0x65, 0x70]);

const bitStringValue = new Uint8Array(1 + keyBytes.length);
bitStringValue[0] = 0x00;
bitStringValue.set(keyBytes, 1);

const bitString = new Uint8Array(2 + bitStringValue.length);
bitString[0] = 0x03;
bitString[1] = bitStringValue.length;
bitString.set(bitStringValue, 2);

const contentLen = algorithmId.length + bitString.length;
const sequence = new Uint8Array(2 + contentLen);
sequence[0] = 0x30;
sequence[1] = contentLen;
sequence.set(algorithmId, 2);
sequence.set(bitString, 2 + algorithmId.length);

return sequence;
}

function getGenerateProperties(): huks.HuksParam[] {
  const properties: huks.HuksParam[] = [
    {
      tag: huks.HuksTag.HUKS_TAG_ALGORITHM,
      value: huks.HuksKeyAlg.HUKS_ALG_ED25519
    },
    {
      tag: huks.HuksTag.HUKS_TAG_PURPOSE,
      value: huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_SIGN | huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_VERIFY
    },
    {
      tag: huks.HuksTag.HUKS_TAG_KEY_SIZE,
      value: huks.HuksKeySize.HUKS_CURVE25519_KEY_SIZE_256
    }
  ];
  return properties;
}

function getSignProperties(): huks.HuksParam[] {
  const properties: huks.HuksParam[] = [
    {
      tag: huks.HuksTag.HUKS_TAG_ALGORITHM,
      value: huks.HuksKeyAlg.HUKS_ALG_ED25519
    },
    {
      tag: huks.HuksTag.HUKS_TAG_PURPOSE,
      value: huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_SIGN
    }
  ];
}

```

```

    return properties;
}

async function generateKey(): Promise<void> {
    const options: huks.HuksOptions = {
        properties: getGenerateProperties()
    };
    try {
        await huks.generateKeyItem(KEY_ALIAS, options);
    } catch (error) {
        throw new Error('Failed to generate device key');
    }
}

async function signData(data: Uint8Array): Promise<Uint8Array> {
    const options: huks.HuksOptions = {
        properties: getSignProperties(),
        inData: data
    };
    try {
        const sessionResult = await huks.initSession(KEY_ALIAS, options);
        const finishResult = await huks.finishSession(sessionResult.handle, options);
        return finishResult.outData as Uint8Array;
    } catch (error) {
        throw new Error('Failed to sign data');
    }
}

async function exportPublicKeyDer(): Promise<Uint8Array> {
    const emptyOptions: huks.HuksOptions = {
        properties: []
    };
    try {
        const result = await huks.exportKeyItem(KEY_ALIAS, emptyOptions);
        return result.outData as Uint8Array;
    } catch (error) {
        throw new Error('Failed to export device public key');
    }
}

export interface DeviceKeyPair {
    publicKeyPem: string;
}

let cachedKeyPair: DeviceKeyPair | null = null;

export async function ensureDeviceKeyPair(): Promise<DeviceKeyPair> {
    if (cachedKeyPair !== null) {
        return cachedKeyPair;
    }
}

```

```

let rawKey: Uint8Array;
try {
  rawKey = await exportPublicKeyDer();
} catch {
  await generateKey();
  rawKey = await exportPublicKeyDer();
}

const spkiDer = buildEd25519SpkiDer(rawKey);
const publicKeyPem = wrapDerInPem(spkiDer);
cachedKeyPair = { publicKeyPem };
return cachedKeyPair;
}

export async function signChallenge(challenge: string): Promise<string> {
  const data = stringToUint8Array(challenge);
  const sig = await signData(data);
  return uint8ArrayToBase64(sig);
}

import { huks } from '@kit.UniversalKeystoreKit';
import { buffer, util } from '@kit.ArkTS';

const CONNECTION_KEY_ALIAS: string = 'codex_router_connection_store_key';
const ENCRYPTED_STORE_PREFIX: string = 'ENC1: ';
const GCM_TAG_LENGTH: number = 16;
const GCM_NONCE_LENGTH: number = 12;

function connectionKeyParams(): huks.HuksParam[] {
  return [
    { tag: huks.HuksTag.HUKS_TAG_ALGORITHM, value: huks.HuksKeyAlg.HUKS_ALG_AES },
    { tag: huks.HuksTag.HUKS_TAG_KEY_SIZE, value: huks.HuksKeySize.HUKS_AES_KEY_SIZE_256 },
    {
      tag: huks.HuksTag.HUKS_TAG_PURPOSE,
      value: huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_ENCRYPT | huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_DECRYPT
    },
    { tag: huks.HuksTag.HUKS_TAG_PADDING, value: huks.HuksKeyPadding.HUKS_PADDING_NONE },
    { tag: huks.HuksTag.HUKS_TAG_BLOCK_MODE, value: huks.HuksCipherMode.HUKS_MODE_GCM }
  ];
}

function cipherParams(purpose: huks.HuksKeyPurpose, nonce: Uint8Array | null, tag: Uint8Array | null):
huks.HuksParam[] {
  const params: huks.HuksParam[] = [
    { tag: huks.HuksTag.HUKS_TAG_ALGORITHM, value: huks.HuksKeyAlg.HUKS_ALG_AES },
    { tag: huks.HuksTag.HUKS_TAG_KEY_SIZE, value: huks.HuksKeySize.HUKS_AES_KEY_SIZE_256 },
    { tag: huks.HuksTag.HUKS_TAG_PURPOSE, value: purpose },
    { tag: huks.HuksTag.HUKS_TAG_PADDING, value: huks.HuksKeyPadding.HUKS_PADDING_NONE },
    { tag: huks.HuksTag.HUKS_TAG_BLOCK_MODE, value: huks.HuksCipherMode.HUKS_MODE_GCM }
  ];
}

```

```

    if (nonce !== null) {
      params.push({ tag: huks.HuksTag.HUKS_TAG_NONCE, value: nonce });
    }

    if (tag !== null) {
      params.push({ tag: huks.HuksTag.HUKS_TAG_AE_TAG, value: tag });
    }

    return params;
  }

  async function ensureConnectionKey(): Promise<void> {
    try {
      await huks.generateKeyItem(CONNECTION_KEY_ALIAS, { properties: connectionKeyParams() });
    } catch (error) {
    }
  }

  function utf8Encode(value: string): Uint8Array {
    const textEncoder = new util.TextEncoder();
    return textEncoder.encodeInto(value);
  }

  function utf8Decode(value: Uint8Array): string {
    const textDecoder = new util.TextDecoder('utf-8');
    return textDecoder.decodeToString(value);
  }

  function copyBufferToBytes(value: buffer.Buffer): Uint8Array {
    const bytes = new Uint8Array(value.length);
    for (let i = 0; i < value.length; i++) {
      bytes[i] = value[i];
    }
    return bytes;
  }

  export function isEncryptedConnectionStore(storedValue: string): boolean {
    return storedValue.startsWith(ENCRYPTED_STORE_PREFIX);
  }

  export async function encryptConnectionStore(plainText: string): Promise<string> {
    try {
      const options: huks.HuksOptions = {
        properties: cipherParams(huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_ENCRYPT, null, null),
        inData: utf8Encode(plainText)
      };

      await ensureConnectionKey();
      const handle = await huks.initSession(CONNECTION_KEY_ALIAS, options);
      const finish = await huks.finishSession(handle.handle, options);
    }
  }

```

```

    const outData = finish.outData;

    if (outData === undefined || outData.length === 0) {
        throw new Error('Failed to encrypt connection store');
    }

    return ENCRYPTED_STORE_PREFIX + buffer.from(outData).toString('base64');
} catch (error) {
    throw new Error('Failed to encrypt connection store');
}
}

export async function decryptConnectionStore(storedValue: string): Promise<string | null> {
    if (!isEncryptedConnectionStore(storedValue)) {
        return null;
    }

    let blob: Uint8Array;
    try {
        const decoded = buffer.from(storedValue.substring(ENCRYPTED_STORE_PREFIX.length), 'base64');
        blob = copyBufferToBytes(decoded);
    } catch (error) {
        return null;
    }

    const minimumLength: number = GCM_NONCE_LENGTH + GCM_TAG_LENGTH;
    if (blob.length < minimumLength) {
        return null;
    }

    const nonce = blob.subarray(blob.length - GCM_NONCE_LENGTH);
    const tag = blob.subarray(blob.length - GCM_NONCE_LENGTH - GCM_TAG_LENGTH, blob.length - GCM_NONCE_LENGTH);
    const cipher = blob.subarray(0, blob.length - GCM_NONCE_LENGTH - GCM_TAG_LENGTH);
    const options: huks.HuksOptions = {
        properties: cipherParams(huks.HuksKeyPurpose.HUKS_KEY_PURPOSE_DECRYPT, nonce, tag),
        inData: cipher
    };

    try {
        const handle = await huks.initSession(CONNECTION_KEY_ALIAS, options);
        const finish = await huks.finishSession(handle.handle, options);
        const outData = finish.outData;

        if (outData === undefined || outData.length === 0) {
            return null;
        }

        return utf8Decode(outData);
    } catch (error) {
        return null;
    }
}

```

```

    }
  }
  export type ConnectionStatus = 'online' | 'offline' | 'needs_login' | 'unknown';
  export type ConnectionSwitchIntent = 'turn_on_and_show_resumes' | 'turn_off' | 'require_login';
  export type ResumeFeedbackAction = 'none' | 'relogin';
  export type ConnectionAgent = 'OpenCode';

  export interface SavedConnection {
    id: string;
    name: string;
    agent: ConnectionAgent;
    host: string;
    port: number;
    scheme: string;
    username: string;
    lastUsedAt: string;
    status: ConnectionStatus;
    accessTokenRef: string;
    refreshTokenRef: string;
    privateKeyRef: string;
  }

  export interface NormalizedServerAddress {
    scheme: string;
    host: string;
    port: number;
    baseUrl: string;
  }

  export interface EditableConnectionFields {
    id: string;
    name: string;
    agent: ConnectionAgent;
    host: string;
    port: number;
    scheme: string;
    username: string;
    lastUsedAt: string;
    accessTokenRef?: string;
    refreshTokenRef?: string;
    privateKeyRef?: string;
  }

  export interface ConnectionFormErrors {
    address: string;
    privateKey: string;
    hasError: boolean;
  }

  export interface ConnectionCarouselSelection {

```

```

    index: number;
    id: string;
}

// 固定单服务器: 只允许一条连接。
export const MAX_CONNECTIONS: number = 1;

const DEFAULT_SCHEME: string = 'https';
const DEFAULT_PORT: number = 8443;
const RESUME_CONTENT_BASE_FONT_SCALE: number = 0.9;

export function normalizeServerAddress(input: string): NormalizedServerAddress {
    let trimmedInput = input.trim();
    let scheme = DEFAULT_SCHEME;
    let address = trimmedInput;

    const schemeSeparatorIndex = trimmedInput.indexOf('://');
    if (schemeSeparatorIndex >= 0) {
        scheme = trimmedInput.substring(0, schemeSeparatorIndex);
        address = trimmedInput.substring(schemeSeparatorIndex + 3);
    }

    let host = address;
    let port = DEFAULT_PORT;
    const portSeparatorIndex = address.lastIndexOf(':');
    if (portSeparatorIndex > -1) {
        const portText = address.substring(portSeparatorIndex + 1);
        const parsedPort = Number(portText);
        if (parsedPort === parsedPort && parsedPort > 0 && parsedPort <= 65535) {
            host = address.substring(0, portSeparatorIndex);
            port = parsedPort;
        }
    }

    return {
        scheme: scheme,
        host: host,
        port: port,
        baseUrl: scheme + '://' + host + ':' + port.toString()
    };
}

export function resolveConnectionName(
    customName: string,
    sessionName: string,
    username: string,
    host: string
): string {
    const normalizedCustomName = customName.trim();
    if (normalizedCustomName.length > 0) {

```

```

    return normalizedCustomName;
}

const normalizedHost = host.trim();
if (normalizedHost.length > 0) {
    return normalizedHost;
}

const normalizedSessionName = sessionName.trim();
if (normalizedSessionName.length > 0) {
    return normalizedSessionName;
}

return username.trim();
}

export function resolveConnectionDisplayName(
    connections: SavedConnection[],
    id: string,
    fallbackName: string
): string {
    const found = connections.find((item: SavedConnection) => item.id === id);
    if (found !== undefined) {
        return found.name;
    }

    return fallbackName;
}

export function resolveConnectionCardTitle(connection: SavedConnection): string {
    return connection.host.trim();
}

export function canAddConnection(connections: SavedConnection[]): boolean {
    return connections.length < MAX_CONNECTIONS;
}

function normalizeConnectionIp(ip: string): string {
    return ip.trim().toLowerCase();
}

export function generateSavedConnectionId(ip: string): string {
    return `OpenCode-${normalizeConnectionIp(ip)}`;
}

export function hasConnectionWithSameEndpoint(
    connections: SavedConnection[],
    scheme: string,
    host: string,
    port: number,

```

```

    ignoredConnectionId: string
  ): boolean {
    const targetId = generateSavedConnectionId(host);

    return connections.some((item: SavedConnection) => {
      if (ignoredConnectionId.length > 0 && item.id === ignoredConnectionId) {
        return false;
      }

      return generateSavedConnectionId(item.host) === targetId;
    });
  }

export function sortConnectionsByLastUsed(connections: SavedConnection[]): SavedConnection[] {
  return connections.slice().sort((left: SavedConnection, right: SavedConnection) => {
    return Date.parse(right.lastUsedAt) - Date.parse(left.lastUsedAt);
  });
}

export function resolveCarouselIndex(currentIndex: number, itemCount: number): number {
  if (itemCount <= 0) {
    return 0;
  }

  if (currentIndex < 0) {
    return 0;
  }

  if (currentIndex >= itemCount) {
    return itemCount - 1;
  }

  return currentIndex;
}

export function resolveCarouselSwipeIndex(currentIndex: number, itemCount: number, offsetX: number): number {
  if (itemCount <= 1) {
    return 0;
  }

  const threshold = 24;
  if (offsetX < -threshold) {
    return currentIndex >= itemCount - 1 ? 0 : currentIndex + 1;
  }

  if (offsetX > threshold) {
    return currentIndex <= 0 ? itemCount - 1 : currentIndex - 1;
  }

  return resolveCarouselIndex(currentIndex, itemCount);
}

```

```

    cwd: item.subtitle,
    status: running ? 'running' : item.status,
    turns
  });
}

// 会把文件写入磁盘的工具（read 之类的只读工具不计入）。
const FILE_PRODUCING_TOOLS = new Set([
  'write', 'edit', 'patch', 'apply_patch', 'multiedit', 'create', 'write_file', 'str_replace_editor'
]);

// shell 类工具：需从命令/输出里扫文件路径（bash 直接写文件不会被上面识别）。
const SHELL_TOOLS = new Set(['bash', 'shell', 'sh', 'console', 'terminal']);

// 会话兜底命名：取首条消息前 N 个字符。
const SESSION_TITLE_MAX_LENGTH = 20;

// 会被登记的产出文件扩展名（与 file-service 白名单一致）。
/**
 * 从会话消息 parts 中提取“产出/改动的文件路径”。
 * 用于把文件按会话归属到设备（不依赖 opencode 的目录隔离）。
 *
 * 覆盖：
 * - `file` part 的 filename/path
 * - 写文件类工具（write/edit/patch...）input/metadata 中含 file/path 的字段
 * - bash/shell 工具的命令与输出中以白名单扩展名结尾的路径（覆盖 `cat > x.docx`、脚本产出等）
 */
export function extractFilePathsFromMessages(messages) {
  if (!Array.isArray(messages)) {
    return [];
  }

  const paths = new Set();
  const add = (value) => {
    const path = normalizeString(value);
    if (path.length > 0) {
      paths.add(path);
    }
  };

  for (const message of messages) {
    const parts = Array.isArray(message?.parts) ? message.parts : [];
    for (const part of parts) {
      const type = normalizeString(part?.type);
      if (type === 'file') {
        add(part?.filename);
        add(part?.path);
        continue;
      }
      if (type !== 'tool') {

```

```

    continue;
  }

  const toolName = normalizeString(part?.tool).toLowerCase();
  const state = part?.state ?? {};

  if (FILE_PRODUCING_TOOLS.has(toolName)) {
    // 不同工具/版本字段名不一 (filePath / file_path / path / filename ...)。
    for (const value of pickPathLikeValues(state.input)) {
      add(value);
    }
    for (const value of pickPathLikeValues(state.metadata)) {
      add(value);
    }
  }

  // shell 类工具: 从命令/输出里扫出以白名单扩展名结尾的路径 (覆盖 bash/脚本产物)。
  if (SHELL_TOOLS.has(toolName)) {
    for (const value of collectStringValue(state.input)) {
      for (const token of extractPathTokens(value)) {
        add(token);
      }
    }
    for (const value of collectStringValue(state.metadata)) {
      for (const token of extractPathTokens(value)) {
        add(token);
      }
    }
    for (const token of extractPathTokens(normalizeString(state.output))) {
      add(token);
    }
  }
}

return Array.from(paths);
}

/** 从 `~/session/:id/diff` 的返回中提取文件路径 (字段名未知, 做通用兜底)。 */
export function extractFilePathsFromDiff(diff) {
  if (!Array.isArray(diff)) {
    return [];
  }
  const paths = new Set();
  const add = (value) => {
    const path = normalizeString(value);
    if (path.length > 0) {
      paths.add(path);
    }
  };
};

```

```

for (const entry of diff) {
  if (!entry || typeof entry !== 'object') {
    continue;
  }
  add(entry.file);
  add(entry.path);
  add(entry.filename);
  for (const value of pickPathLikeValues(entry)) {
    add(value);
  }
  for (const value of collectStringValues(entry)) {
    for (const token of extractPathTokens(value)) {
      add(token);
    }
  }
}
return Array.from(paths);
}

/** 收集对象中所有字符串值（一层）。 */
function collectStringValues(source) {
  if (!source || typeof source !== 'object') {
    return [];
  }
  return Object.values(source).filter((value) => typeof value === 'string');
}

/** 从任意文本里提取“以白名单扩展名结尾”的路径 token。 */
function extractPathTokens(text) {
  const value = normalizeString(text);
  if (value.length === 0) {
    return [];
  }
  const tokens = value.split(/[\s"'`|<>;(){}]+/);
  const results = [];
  for (const raw of tokens) {
    const token = raw.replace(/[\.,;]+$/, '');
    if (token.length > 0 && token.length < 1024 && isAllowedFile(token)) {
      results.push(token);
    }
  }
  return results;
}

function pickPathLikeValues(source) {
  if (!source || typeof source !== 'object') {
    return [];
  }
  const values = [];
  for (const [key, value] of Object.entries(source)) {

```

```

    if (typeof value === 'string' && value.length > 0 && value.length < 1024 && /file|path/i.test(key)) {
      values.push(value);
    }
  }
  return values;
}

export class OpenCodeServerClient {
  constructor(options = {}) {
    this.url = stripTrailingSlash(options.url ?? process.env.OPENCODE_SERVER_URL ?? DEFAULT_OPENCODE_SERVER_URL);
    this.command = options.command ?? process.env.OPENCODE_COMMAND ?? DEFAULT_OPENCODE_COMMAND;
    this.workdir = normalizeString(options.workdir ?? process.env.OPENCODE_WORKDIR ?? process.env.HOME ??
homedir()) ||
    process.cwd();
    this.process = null;
    this.fetchImpl = options.fetchImpl ?? fetch;
    this.spawnServer = options.spawnServer ?? spawn;
    this.platform = options.platform ?? process.platform;
    this.runCommand = options.runCommand ?? ((args) => runCommand(this.spawnServer, this.command, args, {
      cwd: this.workdir,
      platform: this.platform
    }));
    this.now = options.now ?? Date.now;
    this.stderr = '';
    this.startupError = null;
    this.sessionsCacheTtlMs = options.sessionsCacheTtlMs ?? SESSIONS_CACHE_TTL_MS;
    this.sessionsCache = new Map();
  }

  async listResumes(options = {}) {
    const limit = options.limit ?? 20;
    const normalizedLimit = Number.isFinite(limit) && limit > 0 ? limit : 20;
    const rows = await this.loadSessionRows(normalizedLimit);
    const sessions = rows.map(mapOpenCodeDbSessionToSession);
    return sessions
      .filter((session) => !session?.time?.archived)
      .sort((left, right) => Number(right?.time?.updated || 0) - Number(left?.time?.updated || 0))
      .slice(0, normalizedLimit)
      .map(mapOpenCodeSessionToResumeItem);
  }

  /** 带短 TTL 缓存地读取会话列表，避免每次请求都 spawn `opencode db`。 */
  async loadSessionRows(limit) {
    const cached = this.sessionsCache.get(limit);
    const now = this.now();
    if (cached && now - cached.at < this.sessionsCacheTtlMs) {
      return cached.rows;
    }
    const rows = await this.querySessions(limit);
    this.sessionsCache.set(limit, { at: now, rows });
    return rows;
  }

```

```

}

invalidateSessionsCache() {
  this.sessionsCache.clear();
}

async readResume(options = {}) {
  const threadId = requireThreadId(options.threadId);
  const directory = normalizeString(options.directory);
  const session = await this.request(`/session/${encodeURIComponent(threadId)}`, { directory });
  const messages = await this.request(`/session/${encodeURIComponent(threadId)}/message`, { directory });
  const messageFiles = extractFilePathsFromMessages(messages);
  // 会话 diff 能覆盖 bash/脚本产出的文件; 失败时静默降级为仅用消息解析结果。
  const diffFiles = await this.readSessionDiff(threadId, directory);
  return {
    ...mapOpenCodeSessionToResumeSession(session, messages),
    files: Array.from(new Set([...messageFiles, ...diffFiles]))
  };
}

async readSessionDiff(threadId, directory) {
  try {
    const diff = await this.request(`/session/${encodeURIComponent(threadId)}/diff`, {
      directory,
      timeout: 5000
    });
    return extractFilePathsFromDiff(diff);
  } catch (error) {
    return [];
  }
}

async startResume(options = {}) {
  // directory 决定 opencode 的项目/工作目录: 不同设备各自的工作区。
  const directory = normalizeString(options.directory) || normalizeString(options.cwd);
  // 不复用目录当标题: 让 opencode 按首句自动命名 (未自动命名时由 sendResumeMessage 兜底)。
  const session = await this.request('/session', {
    method: 'POST',
    body: {},
    directory: directory !== '~' ? directory : ''
  });
  this.invalidateSessionsCache();
  return mapOpenCodeSessionToResumeSession(session, []);
}

async sendResumeMessage(options = {}) {
  const threadId = requireThreadId(options.threadId);
  const message = normalizeString(options.message);
  if (message.length === 0) {
    throw new Error('message is required');
  }
}

```

```

}

// prompt_async: 发送即返回 204, 不等待生成完成; 结果由客户端轮询 /resume 获取 (不阻塞、不取消生成)。
await this.request(`/session/${encodeURIComponent(threadId)}/prompt_async`, {
  method: 'POST',
  body: {
    parts: [
      {
        type: 'text',
        text: message
      }
    ]
  },
  directory: normalizeString(options.directory)
});

const session = await this.readResume({ threadId, directory: options.directory });

// 兜底命名: 若 opencode 还没按首句命名 (标题仍为空/就是 session id), 用本条消息前 20 字命名。
if (isSessionTitleUnset(session.title, threadId)) {
  const title = deriveSessionTitle(message);
  if (title.length > 0) {
    try {
      await this.renameResume({ threadId, name: title, directory: normalizeString(options.directory) });
      session.title = title;
    } catch (error) {
      console.error(`[opencode] 会话兜底命名失败: ${error instanceof Error ? error.message : error}`);
    }
  }
}

return session;
}

async archiveResume(options = {}) {
  const threadId = requireThreadId(options.threadId);
  const now = Number(this.now());
  await this.runCommand([
    'db',
    `update session set time_archived = ${now}, time_updated = ${now} where id = '${escapeSqlLiteral(threadId)}'`
  ]);
  this.invalidateSessionsCache();
}

async renameResume(options = {}) {
  const threadId = requireThreadId(options.threadId);
  const name = normalizeString(options.name);
  if (name.length === 0) {
    throw new Error('name is required');
  }
}

```

```

    await this.request(`/session/${encodeURIComponent(threadId)}`, {
      method: 'PATCH',
      body: {
        title: name
      },
      directory: normalizeString(options.directory)
    });
    this.invalidateSessionsCache();
  }

  async deleteResume(options = {}) {
    const threadId = requireThreadId(options.threadId);
    await this.runCommand(['session', 'delete', threadId]);
    this.invalidateSessionsCache();
  }

  async querySessions(limit) {
    const stdout = await this.runCommand([
      'db',
      `select id, title, slug, directory, path, time_created, time_updated, time_archived from session where
time_archived is null order by time_updated desc limit ${limit}`,
      '--format',
      'json'
    ]);
    const text = normalizeString(stdout);
    return text.length > 0 ? JSON.parse(text) : [];
  }

  async request(path, options = {}) {
    await this.ensureReady();

    // opencode 按 `directory` 解析项目/工作目录; 用它把不同设备隔离到各自工作区。
    let target = `${this.url}${path}`;
    if (normalizeString(options.directory).length > 0) {
      const url = new URL(target);
      url.searchParams.set('directory', normalizeString(options.directory));
      target = url.toString();
    }

    const timeoutMs = options.timeout ?? 300000;
    const response = await this.fetchImpl(target, {
      method: options.method ?? 'GET',
      headers: options.body !== undefined ? {
        'content-type': 'application/json'
      } : undefined,
      body: options.body !== undefined ? JSON.stringify(options.body) : undefined,
      signal: AbortSignal.timeout(timeoutMs)
    });
    if (!response?.ok) {
      throw new Error(`opencode server request failed: ${response?.status ?? 'no response'}`);
    }
  }

```

```

    }

    const text = await response.text();
    return text.trim().length > 0 ? JSON.parse(text) : {};
}

async ensureReady() {
    if (await this.isReady()) {
        return;
    }

    await this.startServer();
    const startedAt = Date.now();
    while (Date.now() - startedAt < SERVER_READY_TIMEOUT_MS) {
        if (await this.isReady()) {
            return;
        }
        if (!this.process) {
            if (this.startupError) {
                throw this.startupError;
            }
            throw new Error(this.describeStartupError('opencode serve exited before ready'));
        }
        await wait(SERVER_READY_INTERVAL_MS);
    }

    throw new Error(this.describeStartupError('Timed out waiting for opencode serve'));
}

async isReady() {
    try {
        const response = await this.fetchImpl(`${this.url}/session`);
        return Boolean(response?.ok);
    } catch (_error) {
        return false;
    }
}

async startServer() {
    if (this.process) {
        return;
    }

    const listenUrl = new URL(this.url);
    const spawnOptions = {
        stdio: ['ignore', 'ignore', 'pipe'],
        cwd: this.workdir
    };
    if (this.platform === 'win32') {
        spawnOptions.shell = true;
    }
}

```

```

    spawnOptions.windowsHide = true;
  }

  const child = this.spawnServer(this.command, [
    'serve',
    '--hostname',
    listenUrl.hostname,
    '--port',
    listenUrl.port || '4096',
    '--print-logs'
  ], spawnOptions);
  this.process = child;
  this.startupError = null;

  child.stderr?.on('data', (chunk) => {
    this.stderr += chunk.toString();
    if (this.stderr.length > 4000) {
      this.stderr = this.stderr.slice(-4000);
    }
  });

  child.once('error', (error) => {
    this.process = null;
    this.startupError = error;
  });

  child.once('exit', () => {
    this.process = null;
  });
}

describeStartupError(message) {
  const stderr = this.stderr.trim();
  if (stderr.length === 0) {
    return message;
  }

  return `${message}\n${stderr}`;
}

function mapOpenCodeDbSessionToSession(row) {
  return {
    id: row?.id,
    title: row?.title,
    slug: row?.slug,
    directory: row?.directory,
    path: row?.path,
    time: {
      created: row?.time_created,

```

```

        updated: row?.time_updated,
        archived: row?.time_archived
    }
    };
}

function mapOpenCodePartToResumeContent(messageId, role, part, partIndex) {
    const partId = normalizeString(part?.id) || `part-${partIndex + 1}`;
    const type = normalizeString(part?.type) || 'part';

    if (type === 'text') {
        if (role !== 'user' && role !== 'assistant') {
            return null;
        }

        return resumeContent(`${messageId}:${partId}`, role, 'message', part?.text, '');
    }

    // 工具调用 / 文件产物也映射为内容项，让“生成文件/调用工具”这类长任务在客户端有进度可看。
    if (type === 'tool') {
        const toolName = normalizeString(part?.tool) || 'tool';
        const status = normalizeString(part?.state?.status);
        const text = status.length > 0 ? `${toolName} • ${status}` : toolName;
        return resumeContent(`${messageId}:${partId}`, 'tool', 'tool', text, status);
    }

    if (type === 'file') {
        const filename = normalizeString(part?.filename) || normalizeString(part?.path) || 'file';
        // 只展示文件名+后缀，不展示完整路径。
        return resumeContent(`${messageId}:${partId}`, 'tool', 'file', baseName(filename), '');
    }

    return null;
}

function resumeContent(id, role, kind, text, status) {
    const normalizedText = truncateText(normalizePartText(text));
    if (normalizedText.length === 0) {
        return null;
    }

    return {
        id,
        role,
        kind,
        text: normalizedText,
        status
    };
}

```

```
function normalizePartText(value) {
  if (typeof value === 'string') {
    return value.trim();
  }

  if (Array.isArray(value)) {
    return value.map(normalizePartText).filter(Boolean).join('\n');
  }

  if (value && typeof value === 'object') {
    return JSON.stringify(value);
  }

  return '';
}

function normalizeRole(role) {
  if (role === 'assistant' || role === 'user' || role === 'tool' || role === 'system') {
    return role;
  }

  return 'system';
}

function resolveMessageStatus(info) {
  if (info?.error) {
    return 'error';
  }

  if (info?.role === 'assistant' && !info?.time?.completed) {
    return 'running';
  }

  return 'completed';
}

function isActiveStatus(status) {
  return status === 'running' || status === 'inProgress' || status === 'in_progress' || status === 'queued';
}

function timestampToIso(value) {
  const timestamp = Number(value || 0);
  return timestamp > 0 ? new Date(timestamp).toISOString() : '';
}

function truncateText(value) {
  if (value.length <= MAX_PART_TEXT_LENGTH) {
    return value;
  }
}
```

```

    return `${value.slice(0, MAX_PART_TEXT_LENGTH)}\n...`;
}

function normalizeString(value) {
    return typeof value === 'string' ? value.trim() : '';
}

/** 只取路径的文件名部分（兼容 `/`、`.`、`.`）。 */
function baseName(value) {
    const text = normalizeString(value).replace(/\\/g, '/');
    const index = text.lastIndexOf('/');
    return index >= 0 ? text.slice(index + 1) : text;
}

/** 会话标题是否“未命名”（空 / 就是 session id / 形如 ses_xxx）。 */
function isSessionTitleUnset(title, threadId) {
    const value = normalizeString(title);
    if (value.length === 0) {
        return true;
    }
    if (value === normalizeString(threadId)) {
        return true;
    }
    return /^ses[_-]/i.test(value);
}

/** 兜底标题：取消息前 N 个字符（换行保留）。 */
function deriveSessionTitle(message) {
    const text = typeof message === 'string' ? message.replace(/\r\n/g, '\n').trim() : '';
    return Array.from(text).slice(0, SESSION_TITLE_MAX_LENGTH).join('');
}

function requireThreadId(threadId) {
    const normalized = normalizeString(threadId);
    if (normalized.length === 0) {
        throw new Error('threadId is required');
    }
}

return normalized;
}

function stripTrailingSlash(value) {
    return normalizeString(value).replace(/\/+$/, '');
}

function escapeSqlLiteral(value) {
    return normalizeString(value).replace(/'/g, '');
}

function runCommand(spawnCommand, command, args, options = {}) {

```

```

return new Promise((resolve, reject) => {
  const spawnOptions = {
    cwd: options.cwd,
    stdio: ['ignore', 'pipe', 'pipe']
  };
  if (options.platform === 'win32') {
    spawnOptions.shell = true;
    spawnOptions.windowsHide = true;
  }

  const child = spawnCommand(command, args, spawnOptions);
  let stdout = '';
  let stderr = '';

  child.stdout?.on('data', (chunk) => {
    stdout += chunk.toString();
  });
  child.stderr?.on('data', (chunk) => {
    stderr += chunk.toString();
  });
  child.once('error', reject);
  child.once('exit', (code) => {
    if (code === 0) {
      resolve(stdout);
      return;
    }
    reject(new Error(`opencode command failed (${code}): ${stderr.trim()}`));
  });
});
}

function wait(ms) {
  return new Promise((resolve) => setTimeout(resolve, ms));
}
const CLEANUP_INTERVAL_MS = 60 * 1000;

export class RateLimiter {
  #windowMs;
  #maxRequests;
  #clients;
  #cleanupTimer;

  constructor(windowMs, maxRequests) {
    this.#windowMs = windowMs;
    this.#maxRequests = maxRequests;
    this.#clients = new Map();

    this.#cleanupTimer = setInterval(() => this.#cleanup(), CLEANUP_INTERVAL_MS);
    if (this.#cleanupTimer.unref) {

```

```

    this.#cleanupTimer.unref();
  }
}

check(key) {
  const now = Date.now();
  const entry = this.#clients.get(key);

  if (!entry || now >= entry.resetAt) {
    this.#clients.set(key, { count: 1, resetAt: now + this.#windowMs });
    return true;
  }

  if (entry.count >= this.#maxRequests) {
    return false;
  }

  entry.count += 1;
  return true;
}

reset(key) {
  this.#clients.delete(key);
}

#cleanup() {
  const now = Date.now();
  for (const [key, entry] of this.#clients) {
    if (now >= entry.resetAt) {
      this.#clients.delete(key);
    }
  }
}

export function getClientIp(request) {
  const forwarded = request.headers['x-forwarded-for'];
  if (forwarded) {
    return forwarded.split(',')[0].trim();
  }

  const realIp = request.headers['x-real-ip'];
  if (realIp) {
    return realIp.trim();
  }

  return request.socket?.remoteAddress ?? '';
}

const FIFTEEN_MINUTES = 15 * 60 * 1000;

```

```

const FIVE_MINUTES = 5 * 60 * 1000;
const ONE_HOUR = 60 * 60 * 1000;

export const registerLimiter = new RateLimiter(FIFTEEN_MINUTES, 5);
export const activateLimiter = new RateLimiter(FIVE_MINUTES, 3);
export const challengeLimiter = new RateLimiter(FIFTEEN_MINUTES, 10);
export const verifyLimiter = new RateLimiter(FIFTEEN_MINUTES, 5);
export const refreshLimiter = new RateLimiter(FIFTEEN_MINUTES, 10);
export const reregisterLimiter = new RateLimiter(ONE_HOUR, 3);
import http from 'node:http';
import https from 'node:https';
import fs from 'node:fs';
import path from 'node:path';
import { randomUUID } from 'node:crypto';
import { OpenCodeServerClient } from './opencode-server.mjs';
import {
  verifyAccessToken,
  ensureDeviceRegistrationCode,
  createActivationChallenge,
  verifyActivation,
  createLoginChallenge,
  verifyLogin,
  refreshAccessToken,
  reregisterDevice,
  logoutDevice,
  cleanupExpiredNonces,
} from './auth.mjs';
import {
  registerLimiter,
  activateLimiter,
  challengeLimiter,
  verifyLimiter,
  refreshLimiter,
  reregisterLimiter,
  getClientIp,
} from './rate-limiter.mjs';
import {
  resolveFilesRoot,
  deviceWorkspace,
  isAllowedFile,
  mimeTypeForFile,
} from './file-service.mjs';
import {
  saveDeviceSession,
  getDeviceSessionOwner,
  listDeviceSessions,
  deleteDeviceSession,
  upsertSessionFile,
  listSessionFileRecords,
} from './db.mjs';

```

```

const DEFAULT_HOST = '0.0.0.0';
const DEFAULT_PORT = 8443;

const PROJECT = {
  id: 'codex-router',
  name: 'AI Router',
  defaultBranch: 'master',
  status: 'ready',
  permissions: ['task:create', 'task:read', 'task:write', 'diff:read', 'approval:review']
};

function jsonResponse(response, statusCode, body) {
  response.writeHead(statusCode, {
    'content-type': 'application/json; charset=utf-8',
    'access-control-allow-origin': '*',
    'access-control-allow-methods': 'GET,POST,DELETE,OPTIONS',
    'access-control-allow-headers': 'authorization,content-type'
  });
  response.end(JSON.stringify(body));
}

function errorResponse(response, statusCode, code, message) {
  jsonResponse(response, statusCode, {
    error: { code, message, requestId: `req_${randomUUID()}` }
  });
}

function readJson(request) {
  return new Promise((resolve, reject) => {
    let raw = '';
    request.setEncoding('utf8');
    request.on('data', (chunk) => { raw += chunk; });
    request.on('end', () => {
      if (raw.trim().length === 0) { resolve({}); return; }
      try { resolve(JSON.parse(raw)); } catch (e) { reject(e); }
    });
    request.on('error', reject);
  });
}

function bearerToken(request) {
  const header = request.headers.authorization ?? '';
  if (!header.startsWith('Bearer ')) return '';
  return header.substring('Bearer '.length);
}

function normalizeResumeCwd(value) {
  const cwd = typeof value === 'string' ? value.trim() : '';
  return cwd.length > 0 ? cwd : '~';
}

```

```

}

function requireAuth(request, response) {
  const token = bearerToken(request);
  if (!token) {
    errorResponse(response, 401, 'token_missing', 'Authorization token required');
    return null;
  }
  const payload = verifyAccessToken(token);
  if (!payload) {
    errorResponse(response, 401, 'token_expired', 'Token is expired or invalid');
    return null;
  }
  return { deviceId: payload.sub };
}

/**
 * 会话归属校验：只有创建该会话的设备才能访问。非归属方返回 404（不泄漏会话是否存在）。
 */
function requireThreadOwnership(response, deviceId, threadId) {
  const owner = getDeviceSessionOwner(threadId);
  if (!owner || owner !== deviceId) {
    errorResponse(response, 404, 'session_not_found', 'Session not found');
    return false;
  }
  return true;
}

/**
 * 把会话产生的文件路径登记到 session_files（按设备归属）。
 * 只登记位于 filesRoot 之下、扩展名白名单、且当前存在的文件。
 */
function recordSessionFiles(deviceId, threadId, paths) {
  if (!deviceId || !threadId || !Array.isArray(paths) || paths.length === 0) {
    return;
  }
  const filesRoot = resolveFilesRoot();
  if (!filesRoot) {
    return;
  }
  const root = path.resolve(filesRoot);
  for (const raw of paths) {
    const candidate = typeof raw === 'string' ? raw.trim() : '';
    if (candidate.length === 0) {
      continue;
    }
    const absolute = path.resolve(path.isAbsolute(candidate) ? candidate : path.join(root, candidate));
    if (absolute !== root && !absolute.startsWith(root + path.sep)) {
      continue;
    }
  }
}

```

```

    if (!isAllowedFile(absolute)) {
        continue;
    }
    let stats;
    try {
        stats = fs.statSync(absolute);
    } catch (error) {
        continue;
    }
    if (!stats.isFile()) {
        continue;
    }
    const name = path.relative(root, absolute).split(path.sep).join('/');
    try {
        upsertSessionFile(threadId, deviceId, absolute, name, stats.size, stats.mtime.toISOString());
    } catch (error) {
        console.error(`[gateway] 登记会话文件失败: ${error instanceof Error ? error.message : error}`);
    }
}
}

// 工作区扫描时跳过的目录（避免遍历依赖/构建产物）。
const SKIPPED_DIRS = new Set([
    'node_modules', '.git', '.svn', '.hg', 'build', '.build',
    'cache', 'dist', 'out', 'target', 'cert', 'keys', 'data'
]);

function walkAllowedFiles(dir, depth, onFile) {
    if (depth < 0) {
        return;
    }
    let entries;
    try {
        entries = fs.readdirSync(dir, { withFileTypes: true });
    } catch (error) {
        return;
    }
    for (const entry of entries) {
        const full = path.join(dir, entry.name);
        if (entry.isDirectory()) {
            if (entry.name.startsWith('.') || SKIPPED_DIRS.has(entry.name)) {
                continue;
            }
            walkAllowedFiles(full, depth - 1, onFile);
        } else if (entry.isFile() && !entry.name.startsWith('.') && isAllowedFile(entry.name)) {
            onFile(full);
        }
    }
}

```

```

/**
 * 汇总某设备可见的文件：已登记（session_files）的文件 ∪ 本设备专属工作区目录下的实际文件。
 * 工作区目录按 sha256(deviceId) 生成，天然按设备隔离；两者都限定在设备范围内。
 * 对外统一用扁平文件名（basename），按 basename 去重。
 */
function collectDeviceFiles(deviceId) {
  const filesRoot = resolveFilesRoot();
  if (!filesRoot) {
    return [];
  }
  const root = path.resolve(filesRoot);
  const byName = new Map();

  const add = (absolute) => {
    if (absolute !== root && !absolute.startsWith(root + path.sep)) {
      return;
    }
    if (!isAllowedFile(absolute)) {
      return;
    }
    let stats;
    try {
      stats = fs.statSync(absolute);
    } catch (error) {
      return;
    }
    if (!stats.isFile()) {
      return;
    }
    const name = path.basename(absolute);
    if (name.length === 0 || byName.has(name)) {
      return;
    }
    byName.set(name, { name, path: absolute, size: stats.size, modifiedAt: stats.mtime.toISOString() });
  };

  // 1) 已登记文件
  for (const row of listSessionFileRecords(deviceId)) {
    add(path.resolve(root, row.name));
  }

  // 2) 本设备专属工作区目录下的实际文件
  const workspace = deviceWorkspace(root, deviceId);
  if (workspace) {
    walkAllowedFiles(workspace, 4, add);
  }

  return Array.from(byName.values()).sort((left, right) => left.name.localeCompare(right.name));
}

```

```

function createGatewayHandler(options = {}) {
  const tasks = new Map();
  const opencodeClient = options.opencodeClient ?? new OpenCodeServerClient();

  setInterval(() => {
    cleanupExpiredNonces();
  }, 60 * 1000).unref();

  return async (request, response) => {
    try {
      const url = new URL(request.url ?? '/', 'http://localhost');

      if (request.method === 'OPTIONS') {
        jsonResponse(response, 204, {});
        return;
      }

      // —— Health ——
      if (request.method === 'GET' && url.pathname === '/health') {
        jsonResponse(response, 200, { status: 'ok', sessionId: 'opencode-main' });
        return;
      }

      // —— Auth: Register (get activation challenge) ——
      if (request.method === 'POST' && url.pathname === '/api/auth/register') {
        const ip = getClientIp(request);
        if (!registerLimiter.check(ip)) {
          errorResponse(response, 429, 'rate_limited', 'Too many registration attempts');
          return;
        }

        const body = await readJson(request);
        const deviceId = typeof body.deviceId === 'string' ? body.deviceId.trim() : '';
        const publicKey = typeof body.publicKey === 'string' ? body.publicKey.trim() : '';

        if (!deviceId || !publicKey) {
          errorResponse(response, 400, 'invalid_request', 'deviceId and publicKey are required');
          return;
        }

        // 开放注册：首次注册时服务器为该设备随机分配注册码并绑定（此后不可更改/换绑）。
        const registrationCode = ensureDeviceRegistrationCode(deviceId);
        const challenge = createActivationChallenge(deviceId, publicKey);
        jsonResponse(response, 200, { ...challenge, registrationCode });
        return;
      }

      // —— Auth: Activate (complete registration with signature) ——
      if (request.method === 'POST' && url.pathname === '/api/auth/activate') {
        const body = await readJson(request);

```

```

const deviceId = typeof body.deviceId === 'string' ? body.deviceId.trim() : '';
const activationToken = typeof body.activationToken === 'string' ? body.activationToken.trim() : '';
const signedChallenge = typeof body.signedChallenge === 'string' ? body.signedChallenge.trim() : '';

if (!deviceId || !activationToken || !signedChallenge) {
  errorResponse(response, 400, 'invalid_request', 'deviceId, activationToken, and signedChallenge are
required');
  return;
}

if (!activateLimiter.check(activationToken)) {
  errorResponse(response, 429, 'rate_limited', 'Too many activation attempts');
  return;
}

const result = await verifyActivation(deviceId, activationToken, signedChallenge);
if (result.error) {
  const statusMap = {
    invalid_token: 400, device_mismatch: 403, token_used: 400,
    token_expired: 400, no_public_key: 500, invalid_signature: 403
  };
  errorResponse(response, statusMap[result.error] ?? 400, result.error, result.error);
  return;
}

jsonResponse(response, 200, result);
return;
}

// —— Auth: Challenge (get login nonce) ——
if (request.method === 'POST' && url.pathname === '/api/auth/challenge') {
  const body = await readJson(request);
  const deviceId = typeof body.deviceId === 'string' ? body.deviceId.trim() : '';

  if (!deviceId) {
    errorResponse(response, 400, 'invalid_request', 'deviceId is required');
    return;
  }

  if (!challengeLimiter.check(deviceId)) {
    errorResponse(response, 429, 'rate_limited', 'Too many challenge requests');
    return;
  }

  const result = createLoginChallenge(deviceId);
  if (!result) {
    errorResponse(response, 401, 'device_not_found', 'Device not registered or revoked');
    return;
  }

  jsonResponse(response, 200, result);
}

```

```

    return;
}

// —— Auth: Verify (login with signature) ——
if (request.method === 'POST' && url.pathname === '/api/auth/verify') {
    const body = await readJson(request);
    const deviceId = typeof body.deviceId === 'string' ? body.deviceId.trim() : '';
    const nonce = typeof body.nonce === 'string' ? body.nonce.trim() : '';
    const signature = typeof body.signature === 'string' ? body.signature.trim() : '';

    if (!deviceId || !nonce || !signature) {
        errorResponse(response, 400, 'invalid_request', 'deviceId, nonce, and signature are required');
        return;
    }

    if (!verifyLimiter.check(deviceId)) {
        errorResponse(response, 429, 'rate_limited', 'Too many verification attempts');
        return;
    }

    const result = await verifyLogin(deviceId, nonce, signature);
    if (result.error) {
        const statusMap = {
            invalid_nonce: 400, device_mismatch: 403, nonce_used: 400,
            nonce_expired: 400, device_not_found: 401, invalid_signature: 403
        };
        errorResponse(response, statusMap[result.error] ?? 400, result.error, result.error);
        return;
    }

    jsonResponse(response, 200, result);
    return;
}

// —— Auth: Refresh ——
if (request.method === 'POST' && url.pathname === '/api/auth/refresh') {
    const body = await readJson(request);
    const refreshToken = typeof body.refreshToken === 'string' ? body.refreshToken.trim() : '';

    if (!refreshToken) {
        errorResponse(response, 400, 'invalid_request', 'refreshToken is required');
        return;
    }

    if (!refreshLimiter.check(refreshToken)) {
        errorResponse(response, 429, 'rate_limited', 'Too many refresh attempts');
        return;
    }

    const result = refreshAccessToken(refreshToken);

```

```

    if (result.error) {
      errorResponse(response, 401, result.error, result.error);
      return;
    }

    jsonResponse(response, 200, result);
    return;
  }

  // —— Auth: Logout ——
  if (request.method === 'POST' && url.pathname === '/api/auth/logout') {
    const body = await readJson(request);
    const refreshToken = typeof body.refreshToken === 'string' ? body.refreshToken.trim() : '';
    if (refreshToken) {
      logoutDevice(refreshToken);
    }
    jsonResponse(response, 200, { ok: true });
    return;
  }

  // —— Auth: Reregister ——
  if (request.method === 'POST' && url.pathname === '/api/auth/reregister') {
    const ip = getClientIp(request);
    if (!reregisterLimiter.check(ip)) {
      errorResponse(response, 429, 'rate_limited', 'Too many reregistration attempts');
      return;
    }

    const body = await readJson(request);
    const deviceId = typeof body.deviceId === 'string' ? body.deviceId.trim() : '';
    const publicKey = typeof body.publicKey === 'string' ? body.publicKey.trim() : '';
    const deviceName = typeof body.deviceName === 'string' ? body.deviceName.trim() : '';
    const mode = typeof body.mode === 'string' ? body.mode.trim() : 'reset';

    if (!deviceId || !publicKey) {
      errorResponse(response, 400, 'invalid_request', 'deviceId and publicKey are required');
      return;
    }

    // 注册码无需输入：沿用该设备首次注册时自动分配的码。
    const result = await reregisterDevice(deviceId, publicKey, deviceName, mode);
    if (result.error) {
      errorResponse(response, 401, result.error, result.error);
      return;
    }

    jsonResponse(response, 200, result);
    return;
  }
}

```

```

// —— Auth: Me ——
if (request.method === 'GET' && url.pathname === '/api/auth/me') {
  const auth = requireAuth(request, response);
  if (!auth) return;
  jsonResponse(response, 200, {
    deviceId: auth.deviceId,
    sessionId: 'opencode-main',
    registrationCode: ensureDeviceRegistrationCode(auth.deviceId)
  });
  return;
}

// —— ICE Config ——
if (request.method === 'GET' && url.pathname === '/api/turn/ice-config') {
  const auth = requireAuth(request, response);
  if (!auth) return;

  const turnHost = process.env.COTURN_HOST ?? '127.0.0.1';
  const turnPort = process.env.COTURN_PORT ?? '3478';
  const turnUser = process.env.COTURN_USER ?? 'codexrouter';
  const turnPass = process.env.COTURN_PASS ?? '';

  const iceServers = [{ url: `stun:${turnHost}:${turnPort}` }];
  if (turnPass.length > 0) {
    iceServers.push({ url: `turn:${turnHost}:${turnPort}`, username: turnUser, credential: turnPass });
  }
  jsonResponse(response, 200, { iceServers });
  return;
}

// —— Projects ——
if (request.method === 'GET' && url.pathname === '/api/projects') {
  const auth = requireAuth(request, response);
  if (!auth) return;
  jsonResponse(response, 200, { items: [PROJECT] });
  return;
}

// —— OpenCode Resumes ——
if (request.method === 'GET' && url.pathname === '/api/opencode/resumes') {
  const auth = requireAuth(request, response);
  if (!auth) return;

  const limit = Number(url.searchParams.get('limit')) ?? '20';
  const allItems = await opencodeClient.listResumes({ limit: 1000 });
  // 归属持久化在 SQLite, gateway 重启后依然只列表本设备会话。
  const deviceThreadIds = new Set(listDeviceSessions(auth.deviceId));
  const items = allItems
    .filter((item) => deviceThreadIds.has(item.id))
    .slice(0, Number.isFinite(limit) && limit > 0 ? limit : 20);

```

```

    jsonResponse(response, 200, { items });
    return;
}

if (request.method === 'POST' && url.pathname === '/api/opencode/resumes') {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const filesRoot = resolveFilesRoot();
    if (!filesRoot) {
        errorResponse(response, 500, 'files_root_unset', 'FILES_ROOT or OPENCODE_WORKDIR is not configured');
        return;
    }

    // 设备专属工作目录: 强制会话 cwd 指向该设备目录, 实现文件天然按设备隔离。
    const workspace = deviceWorkspace(filesRoot, auth.deviceId);
    if (!workspace) {
        errorResponse(response, 400, 'invalid_device', 'Device id is required');
        return;
    }
    try {
        fs.mkdirSync(workspace, { recursive: true });
    } catch (error) {
        errorResponse(response, 500, 'workspace_create_failed', 'Could not create device workspace directory');
        return;
    }

    const cwd = workspace;
    // directory 让 opencode 把该会话的读写限制在设备工作区 (文件隔离的关键)。
    const session = await opencodeClient.startResume({ cwd, directory: workspace });
    if (auth.deviceId && session.threadId) {
        saveDeviceSession(session.threadId, auth.deviceId);
    }
    jsonResponse(response, 200, session);
    return;
}

const opencodeResumeMatch = url.pathname.match(/^\/api\/opencode\/resumes\/([^\s]+)\/resume$/);
if (request.method === 'POST' && opencodeResumeMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const threadId = decodeURIComponent(opencodeResumeMatch[1]);
    if (threadId.trim().length === 0) {
        errorResponse(response, 400, 'invalid_request', 'Thread id is required');
        return;
    }
    if (!requireThreadOwnership(response, auth.deviceId, threadId)) return;
    const directory = deviceWorkspace(resolveFilesRoot(), auth.deviceId) ?? '';
    const session = await opencodeClient.readResume({ threadId, directory });
}

```

```

    recordSessionFiles(auth.deviceId, threadId, session.files);
    jsonResponse(response, 200, session);
    return;
}

const opencodeArchiveResumeMatch = url.pathname.match(/^\/api\/opencode\/resumes\/([^\/]*)\/archive$/);
if (request.method === 'POST' && opencodeArchiveResumeMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const threadId = decodeURIComponent(opencodeArchiveResumeMatch[1]);
    if (threadId.trim().length === 0) {
        errorResponse(response, 400, 'invalid_request', 'Thread id is required');
        return;
    }
    if (!requireThreadOwnership(response, auth.deviceId, threadId)) return;
    await opencodeClient.archiveResume({ threadId });
    deleteDeviceSession(threadId);
    jsonResponse(response, 200, { ok: true });
    return;
}

const opencodeRenameResumeMatch = url.pathname.match(/^\/api\/opencode\/resumes\/([^\/]*)\/name$/);
if (request.method === 'POST' && opencodeRenameResumeMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const threadId = decodeURIComponent(opencodeRenameResumeMatch[1]);
    const body = await readJson(request);
    const name = typeof body.name === 'string' ? body.name.trim() : '';
    if (threadId.trim().length === 0 || name.length === 0) {
        errorResponse(response, 400, 'invalid_request', 'Thread id and name are required');
        return;
    }
    if (!requireThreadOwnership(response, auth.deviceId, threadId)) return;
    const directory = deviceWorkspace(resolveFilesRoot(), auth.deviceId) ?? '';
    await opencodeClient.renameResume({ threadId, name, directory });
    jsonResponse(response, 200, { ok: true });
    return;
}

const opencodeDeleteResumeMatch = url.pathname.match(/^\/api\/opencode\/resumes\/([^\/]*)$/);
if (request.method === 'DELETE' && opencodeDeleteResumeMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const threadId = decodeURIComponent(opencodeDeleteResumeMatch[1]);
    if (threadId.trim().length === 0) {
        errorResponse(response, 400, 'invalid_request', 'Thread id is required');
        return;
    }

```

```

    }
    if (!requireThreadOwnership(response, auth.deviceId, threadId)) return;
    await opencodeClient.deleteResume({ threadId });
    deleteDeviceSession(threadId);
    jsonResponse(response, 200, { ok: true });
    return;
}

const opencodeMessageMatch = url.pathname.match(/^\/api\/opencode\/resumes\/([^\/]+)\/messages$/);
if (request.method === 'POST' && opencodeMessageMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const threadId = decodeURIComponent(opencodeMessageMatch[1]);
    const body = await readJson(request);
    const message = typeof body.message === 'string' ? body.message : '';
    if (threadId.trim().length === 0 || message.trim().length === 0) {
        errorResponse(response, 400, 'invalid_request', 'Thread id and message are required');
        return;
    }
    if (!requireThreadOwnership(response, auth.deviceId, threadId)) return;
    const directory = deviceWorkspace(resolveFilesRoot(), auth.deviceId) ?? '';
    const session = await opencodeClient.sendResumeMessage({ threadId, message, directory });
    recordSessionFiles(auth.deviceId, threadId, session.files);
    jsonResponse(response, 200, session);
    return;
}

// —— Tasks ——
const createTaskMatch = url.pathname.match(/^\/api\/projects\/([^\/]+)\/tasks$/);
if (request.method === 'POST' && createTaskMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const projectId = createTaskMatch[1];
    if (projectId !== PROJECT.id) {
        errorResponse(response, 404, 'project_not_found', 'Project not found');
        return;
    }

    const body = await readJson(request);
    const taskId = `task_${randomUUID()}`;
    const task = {
        id: taskId, taskId, projectId, status: 'queued',
        message: typeof body.message === 'string' ? body.message : '',
        sandbox: typeof body.sandbox === 'string' ? body.sandbox : 'workspace-write',
        networkEnabled: Boolean(body.networkEnabled),
        createdAt: new Date().toISOString()
    };
    tasks.set(taskId, task);
}

```

```

    jsonResponse(response, 200, { taskId, status: task.status });
    return;
}

const taskMatch = url.pathname.match(/^\/api\/tasks\/([^/]+)$/);
if (request.method === 'GET' && taskMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;

    const task = tasks.get(taskMatch[1]);
    if (!task) {
        errorResponse(response, 404, 'task_not_found', 'Task not found');
        return;
    }
    jsonResponse(response, 200, task);
    return;
}

// —— Files: List (本设备可见文件 = 登记文件 ∪ 设备专属工作区文件) ——
if (request.method === 'GET' && url.pathname === '/api/files') {
    const auth = requireAuth(request, response);
    if (!auth) return;
    if (!resolveFilesRoot()) {
        errorResponse(response, 500, 'files_root_unset', 'FILES_ROOT or OPENCODE_WORKDIR is not configured');
        return;
    }
    const items = collectDeviceFiles(auth.deviceId)
        .map((file) => ({ name: file.name, size: file.size, modifiedAt: file.modifiedAt }));
    jsonResponse(response, 200, { items });
    return;
}

// —— Files: Download (仅限本设备可见文件: 登记记录或设备专属工作区) ——
const downloadMatch = url.pathname.match(/^\/api\/files\/(.+)\download$/);
if (request.method === 'GET' && downloadMatch) {
    const auth = requireAuth(request, response);
    if (!auth) return;
    if (!resolveFilesRoot()) {
        errorResponse(response, 500, 'files_root_unset', 'FILES_ROOT or OPENCODE_WORKDIR is not configured');
        return;
    }
}

let requestedName;
try {
    requestedName = decodeURIComponent(downloadMatch[1]);
} catch (error) {
    requestedName = downloadMatch[1];
}

// 在“本设备可见文件”里按扁平文件名(basename)匹配, 限定设备范围, 杜绝越权/穿越。

```

```

const wanted = path.basename(requestedName);
const match = collectDeviceFiles(auth.deviceId).find((file) => file.name === wanted);
if (!match) {
  errorResponse(response, 404, 'file_not_found', 'File does not exist');
  return;
}
const target = match.path;
if (!isAllowedFile(target)) {
  errorResponse(response, 400, 'invalid_file', 'File name is invalid or not allowed');
  return;
}

let stats;
try {
  stats = fs.statSync(target);
} catch (error) {
  errorResponse(response, 404, 'file_not_found', 'File does not exist');
  return;
}
if (!stats.isFile()) {
  errorResponse(response, 404, 'file_not_found', 'File does not exist');
  return;
}

// 流式发送, 避免大文件整块读入内存。
response.writeHead(200, {
  'content-type': mimeTypeForFile(target),
  'content-length': stats.size,
  'content-disposition': `attachment; filename="${encodeURIComponent(path.basename(target))}"`
});
const stream = fs.createReadStream(target);
stream.on('error', () => {
  response.destroy();
});
response.on('close', () => {
  stream.destroy();
});
stream.pipe(response);
return;
}

errorResponse(response, 404, 'not_found', 'Not found');
} catch (error) {
  console.error(`[gateway] ${request.method} ${request.url} 500:`, error);
  errorResponse(response, 500, 'gateway_unavailable', error instanceof Error ? error.message : 'Gateway error');
}
}
}

export function createGatewayListener(options = {}) {

```

```

const handler = createGatewayHandler(options);
if (options.tlsKey && options.tlsCert) {
  const createSecureServer = options.createSecureServer ?? https.createServer;
  return createSecureServer({ key: options.tlsKey, cert: options.tlsCert }, handler);
}
return http.createServer(handler);
}

export function createGatewayServer(options = {}) {
  return createGatewayListener(options);
}

export function startGatewayServer(options = {}) {
  const host = options.host ?? process.env.GATEWAY_HOST ?? DEFAULT_HOST;
  const port = Number(options.port ?? process.env.GATEWAY_PORT ?? DEFAULT_PORT);
  const httpPort = Number(options.httpPort ?? process.env.GATEWAY_HTTP_PORT ?? 8080);
  const tlsKeyPath = options.tlsKeyPath ?? process.env.GATEWAY_TLS_KEY;
  const tlsCertPath = options.tlsCertPath ?? process.env.GATEWAY_TLS_CERT;
  const tlsOptions = tlsKeyPath && tlsCertPath ? {
    tlsKey: fs.readFileSync(tlsKeyPath),
    tlsCert: fs.readFileSync(tlsCertPath)
  } : {};
  const handler = createGatewayHandler(options);

  const servers = [];

  if (tlsKeyPath && tlsCertPath) {
    const httpsServer = https.createServer({ key: tlsOptions.tlsKey, cert: tlsOptions.tlsCert }, handler);
    httpsServer.listen(port, host, () => {
      const address = httpsServer.address();
      const actualPort = typeof address === 'object' && address ? address.port : port;
      console.log(`AI Router Gateway listening on https://${host}:${actualPort}`);
    });
    servers.push(httpsServer);
  }

  const httpServer = http.createServer(handler);
  httpServer.listen(httpPort, host, () => {
    const address = httpServer.address();
    const actualPort = typeof address === 'object' && address ? address.port : httpPort;
    console.log(`AI Router Gateway listening on http://${host}:${actualPort}`);
  });
  servers.push(httpServer);

  return servers;
}

if (import.meta.url === `file://${process.argv[1]}`) {
  startGatewayServer();
}

```